Predicted densities and weights in the historical CH instrument

1. Put the historical information on modern county units. Let P_c be historical population assigned to modern county c by geographic overlap. Let n_c and a_c be modern employment and land area in square kilometres. Counties combined in the employment data are kept together. Two historical density definitions are available: ℓ_c is either the population-weighted mean of historical source log densities within c, or $\log(P_c/a_c)$. The first preserves historical source densities. The second spreads assigned population over the full modern county land area. These are different measures.

2. Predict modern employment density from counties. For each historical year and territory scope, estimate by ordinary least squares

$$\log(n_c/a_c) = \alpha + \pi\ell_c + X_c'\gamma + v_c, \qquad \widehat{m}_c = \widehat{\alpha} + \widehat{\pi}\ell_c + X_c'\widehat{\gamma}.$$

Here X_c contains the selected geography and transport variables. The regression includes an intercept and no fixed effects or human-capital terms. It predicts employment density; the county productivity regression is separate. Fitted log density $\widehat{m}_c$ and predicted employment $\widehat{n}_c = a_c \exp(\widehat{m}_c)$ do not depend on θ.

3. Choose what the prediction replaces. For a modern MSA or state g, the derivative instruments are

$$\text{Predicted density:} \quad z_g = \frac{\sum_{c\in g} P_c \widehat{m}_c}{\sum_{c\in g} P_c},$$

$$\text{Predicted employment weights:} \quad z_g = \frac{\sum_{c\in g} \widehat{n}_c \ell_c}{\sum_{c\in g} \widehat{n}_c},$$

$$\text{Predicted density and weights:} \quad z_g = \frac{\sum_{c\in g} \widehat{n}_c \widehat{m}_c}{\sum_{c\in g} \widehat{n}_c}.$$

All three use the same counties. Under the NLS-based method, put the corresponding densities and normalized weights into $H_g(\widehat{\theta}_{\text{NLS}})$ and hold the construction value fixed during GMM. The original historical option can retain more counties when prediction inputs are missing.

4. Coverage and estimation. Prediction requires positive historical population, defined density and complete selected geography. Training also requires positive modern employment. The same national training sample is used for MSAs and states; DC can receive a prediction but does not enter training. Coverage counts known historical area in contributing counties, divided by the region's full mapped area. Zero-population counties cannot enter the prediction. Choose 80%, 95%, or no cutoff for both historical years. The cutoff filters regions after the county regression. Predicted weights use full modern county land area.

Inference conditions on the fitted prediction coefficients and, when selected, the NLS construction value. It excludes their estimation uncertainty. The prediction does not establish the exclusion restriction.